4-38.

(1) $c^2 \nabla \times \vec{B}_1 = \frac{\vec{j}_1}{\varepsilon_0} + \dot{\vec{E}}_1$, $\longrightarrow$ $c^2 \nabla \times \dot{\vec{B}}_1 = \frac{1}{\varepsilon_0} \frac{\partial}{\partial t} \vec{j}_1 + \ddot{\vec{E}}_1$

$\nabla \times \vec{E}_1 = -\dot{\vec{B}}_1$ $\longrightarrow$ $\nabla \times (\nabla \times \vec{E}_1) = -\nabla \times \dot{\vec{B}}_1$

$= \nabla(\nabla \cdot \vec{E}_1) - \nabla^2 \vec{E}$

$\Rightarrow$ $-\vec{k}(\vec{k} \cdot \vec{E}) + k^2 \vec{E} = \frac{i\omega}{\varepsilon_0 c^2} \vec{j} + \frac{\omega^2}{c^2} \vec{E}_1$

对横波: $\vec{k} \cdot \vec{E} = 0$.

$$k^2 \vec{E} = \frac{i\omega}{\varepsilon_0 c^2} \vec{j}_1 + \frac{\omega^2}{c^2} \vec{E}_1$$

$$\left(k^2 - \frac{\omega^2}{c^2}\right) \vec{E} = \frac{i n_0 e \omega \vec{v}_{e2}}{\varepsilon_0}$$

动力学方程:

$$m \frac{\partial \vec{v}_{e2}}{\partial t} - i\omega m \vec{v}_{e2} = -e\vec{E}_2 - m\nu \vec{v}_{e2}$$

碰撞频率.

$$\vec{v}_{e2} = \frac{e\vec{E}_2}{i\omega m + m\nu}$$

$$\omega^2 - c^2 k^2 = \frac{i n_0 e^2 \omega}{\varepsilon_0 (i\omega m + m\nu)}$$

$\Leftrightarrow$ $$\frac{c^2 k^2}{\omega^2} = 1 - \frac{\omega_p^2}{\omega^2 - i\omega\nu}$$ (∇ $n\nu$)



(2) $\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2}{\omega(\omega+i\nu)}$ $\qquad \frac{\nu}{\omega} = \chi$, $\nu = \omega\chi$.

$\therefore$ $$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2}{\omega^2 + i\chi\omega^2}$$

$$\omega^2 = c^2k^2 + \frac{\omega_p^2}{1+i\chi}$$

$$\omega^2 = c^2k^2 + \frac{\omega_p^2}{1+\chi^2}(1-i\chi)$$

设 $\omega = a + ib$, $\omega^2 = a^2 - b^2 + i2ab = c^2k^2 + \frac{\omega_p^2}{1+\chi^2} - i\frac{\omega_p^2\chi}{1+\chi^2}$

$\therefore$ $$\begin{cases} a^2 - b^2 = c^2k^2 + \frac{\omega_p^2}{1+\chi^2} \\ 2ab = -\frac{\omega_p^2\chi}{1+\chi^2} \end{cases}$$

$a = -\frac{\omega_p^2\chi}{2b}$

$\Rightarrow$ $$b^4 + (c^2k^2 + \omega_p^2)b^2 - \frac{\omega_p^4\chi^2}{4} = 0$$

解得: $b^2 = \frac{\omega_p^4\chi^2}{4(c^2k^2+\omega_p^2)}$, $b = \frac{-\omega_p^2\chi}{2\sqrt{c^2k^2+\omega_p^2}} = \frac{-\omega_p^2\nu}{2\omega^2}$

时间阻尼，衰减速率：

$$\gamma \equiv -\mathrm{Im}(\omega) = \frac{\omega_p^2\,\mathrm{Re}(\nu)}{2\,\mathrm{Re}(\omega)}$$

(3) 同样地可求 k.

$$Re[k] =$$

$$Im[k] = \frac{\omega_p^2 \chi}{2c\sqrt{\omega^2-\omega_p^2}} = \frac{Re(\nu)}{2c}\frac{\omega_p^2}{Re(\omega^2)\sqrt{1-\omega_p^2/\omega^2}}$$

空间阻尼:

$$\delta \equiv [Im(k)]^{-1} = \frac{2c}{\omega_p^2}\frac{Re(\omega^2)}{Re(\nu)}\sqrt{1-\frac{\omega_p^2}{\omega^2}}$$